

Roll No.

TCS-549

B. TECH. (CSE) (FIFTH SEMESTER)

MID SEMESTER EXAMINATION, Oct., 2022

DATA SCIENCE VISUALIZATION WITH TABLEAU SPECIALIZATION

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each question carries 10 marks.

1. (a) What is Data Visualization ? Why is data visualization is important ?
Explain its role with appropriate case study. (CO1)

OR

- (b) What is Big Data Visualization ? What are the various challenges in Big Data Visualization ? Explain its role with appropriate case study. (CO1)
2. (a) WASM is a product tailored specifically for social media analysis — pulling in social media content from a number of sources and surfacing insights around everything from trending topics, to sentiment analysis, key influencers, audience demographics etc. Design the methodology for data visualization of WASM. (CO2)

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OR

- (b) Dream Housing Finance that deals in all home loans. They have presence across all urban, semi urban and rural areas. Customer first applies for home loan after that company validates the customer eligibility for loan. However doing this manually takes a lot of time, hence it wants to automate the loan eligibility process based on customer information. Design the methodology to prepare and familiarize of data. (CO2)

3. (a) Banking Frontiers is an exclusive B2B magazine with a closed circulation of 7,500 + and readership of 22,500 + copies. Their readers form a focused segment of industry leaders and key decision makers from this universe. Their stories cover key areas in BFSI like Core Banking. Payments. Treasury, Compliance, AML and Banking innovations. Analyse the importance of Editorial Focus in the perspective of Banking Frontiers. (CO2)

OR

- (b) Bookimed, a Ukrainian service for searching the best medical solutions worldwide, wanted to make an x2 increase in revenue by year. To do so, the companies that use visualization tool would have to make reasonable decisions based on data. They understood that previously managers had issues with evaluating hypothesis and tasks prioritization because of the manual filtering through the information. Refine the editorial focus of visualization tool in the perspective of Bookimed. (CO2)

4. (a) Coca-Cola, the largest beverage company in the world, looked how to replace its daily 45-minute manual data reporting process. Previously, the team spent a considerable amount of time trying to connect over 200 million lines of data from over 100 different systems into single storage to then build one usable dashboard. To boost the efficiency and carry out a real-time data, Coca-Cola adopted Tableau. Compare and contrast Traditional BI Tools and Tableau. (CO2)

OR

- (b) LinkedIn, a largest professional networking website, wanted to synchronize all the data across its internal databases (Google Analytics, Salesforce.com, third-party tools). Previously, one analyst at LinkedIn would handle daily sales request from over 500 salespersons. This created a reporting queue of up to 6 months. To fix the issue, LinkedIn decided to use Tableau to centralize the spread out data and develop a series of customer access dashboards. As a result, thousands of individuals nowadays can access the Tableau Server on a weekly basis, which constitutes 90% of the LinkedIn sales force. With the interactive real time dashboards in Tableau, one can easily predict churn and track the current performance, which eventually created more revenue through the proactive cycle of sales. State some ways to improve the performance of Tableau in the perspective of LinkedIn. (CO2)
5. (a) *In the past two years of the Pandemic, the power of data science has been more evident than ever. Different pharmaceutical companies across the globe could synthesize Covid 19 vaccines by analyzing the data to understand the trends and patterns of the outbreak. Identify the*

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Charts/Graphs/Maps those are useful to track the virus in real-time, predict patterns, devise effective strategies to fight the Pandemic. (CO1)

OR

- (b) Cricket is one of the most popular sports in the world with an estimated fan following of 2.5 billion people. There are primarily three types of international cricket matches—test matches, one day internationals, and Twenty20. Among these, one day internationals (ODIs) are the most commonly played type of matches in international tournaments including the ICC Cricket World Cup and the ICC Champions Trophy. Choose the optimal Graph/Chart to visualize the score of each player and justify the reason to choose. (CO1)